

CS6102 Assignment 2: The Hardware Society Incident

Overview

Welcome to Assignment 2 of CS6102, titled “**The Hardware Society Incident**” (intentionally spelled *Hardwaare*, inspired by its Hindi pronunciation).

In this assignment, you will explore practical side-channel analysis techniques and apply them to recover sensitive information from leakage traces. The goal is to understand how physical implementations of cryptographic systems can leak information, and how attackers can exploit such leakage.

The main interface for this assignment is available at:

<http://10.130.172.141:8000/>

Correlation Power Analysis (CPA)

Correlation Power Analysis (CPA) is a statistical side-channel attack that exploits the relationship between power consumption and intermediate values in a cryptographic algorithm.

The key idea is:

- Assume a leakage model (commonly Hamming Weight or Hamming Distance).
- For each key guess, compute the hypothetical leakage.
- Compare it with actual measured traces using correlation.

Mathematically, CPA computes the Pearson correlation coefficient between:

- Hypothetical leakage vector H
- Measured power traces T

$$\rho(H, T) = \frac{\text{Cov}(H, T)}{\sigma_H \sigma_T} \quad (1)$$

The key guess that produces the highest correlation is considered the correct one.

Template Attacks

Template attacks are a more powerful class of side-channel attacks based on statistical modeling.

Unlike CPA, which uses a simple leakage model, template attacks:

- Use **profiling traces** from a controlled device.
- Build a probabilistic model (typically Gaussian) for each intermediate value.

- During the attack phase, evaluate the likelihood of observed traces under each model.

Each class (e.g., Hamming weight) is modeled using:

- Mean vector μ
- Covariance matrix Σ

The likelihood is computed as:

$$P(T \mid k) \propto \exp \left(-\frac{1}{2} (T - \mu_k)^T \Sigma_k^{-1} (T - \mu_k) \right) \quad (2)$$

Why Template Attacks are Better than CPA

Template attacks improve upon CPA in several ways:

- **Stronger leakage modeling:** They capture full statistical behavior instead of relying on simple models.
- **Higher accuracy:** Effective even with fewer traces.
- **Noise resilience:** Explicitly model noise using covariance.
- **Optimality:** Under Gaussian assumptions, template attacks are statistically optimal.

In contrast, CPA:

- Assumes linear leakage
- Ignores noise structure
- Requires more traces

Provided Files

You are given the following files in the folder:

`cs6102_assignment2`

- `profiling_traces.npz` Contains labeled traces used to build templates. “
- `template_attack.py`
A starting point for implementing the template attack. “

Accessing Attack Traces

- Teams and IDs have already been announced.
- Use your **Team ID** and **password** to log in to the website.
- Download your assigned attack traces from:

`http://10.130.172.141:8000/`

Checking if key is correct

You can check if your attack is successful (i.e. you have extracted the key) by typing the aes key when prompted - (16 integers separated by commas)

like : 1,2,3,4,5,6,7,8,9,0,1,2,3,4,5

Submission Deadline

Deadline: April 10th

Late submissions may not be accepted.

Final Notes

This assignment is designed to give you hands-on experience with real-world side-channel attacks. You are encouraged to:

- Understand the difference between profiling and attack phases
- Experiment with different leakage models
- Optimize your implementation

Good luck, and enjoy breaking (and understanding) hardware security!